

VLADIKAVKAZ, Russian Federation

WMO: 372280

Lat: 43.033N

Lon: 44.683E

Elev: 2306

StdP: 13.51

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	12.5	2.1	6.7	10.6	6.5	8.4	17.2	16.0	39.7	12.1	32.3	2.8	170	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	88.0	68.6	84.8	67.7	81.9	66.7	71.7	83.1	70.1	80.7	68.5	78.5	4.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	112.5	78.1	66.4	105.9	76.2	65.0	100.7	74.2	37.1	83.1	35.6	80.7	34.2	78.6	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	10.0	8.2	DB	2.0	92.8	6.1	4.5	-2.4	96.0	-6.0	98.7	-9.4	101.2	-13.9	104.4	(4)
(5)			WB	1.6	74.7	5.8	2.9	-2.6	76.8	-6.0	78.5	-9.3	80.1	-13.5	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	29.4	31.7	39.7	49.6	59.2	65.9	70.7	70.3	61.2	51.1	39.9	32.1		
	DBStd	16.58	8.22	9.41	8.39	8.46	7.17	5.86	5.09	5.72	6.64	8.25	8.48	8.90		
	HDD50	2575	640	515	336	109	11	0	0	0	5	85	318	556		
	HDD65	5934	1105	934	786	467	203	57	9	18	146	437	753	1019		
	CDD50	2635	0	2	17	96	296	477	640	631	342	117	15	2		
	CDD65	517	0	0	1	3	23	84	184	184	33	5	0	0		
	CDH74	3805	0	0	9	43	189	597	1299	1368	249	51	0	0		
	CDH80	1064	0	0	2	10	32	136	382	432	61	9	0	0		
(14) Wind	WSAvg	3.4	3.1	3.5	4.2	3.5	3.7	3.7	3.5	3.3	3.1	2.7	3.0	3.1		
(15) Precipitation	PrecAvg	35.2	1.1	1.1	1.9	3.1	5.4	6.2	4.8	3.7	2.8	2.0	1.7	1.3		
	PrecMax	56.6	3.0	3.3	4.7	5.9	10.3	9.6	13.7	14.9	5.7	5.4	4.8	4.6		
	PrecMin	24.2	0.1	0.2	0.2	0.1	1.5	3.5	0.5	0.9	0.5	0.0	0.4	0.4		
	PrecStd	7.0	0.7	0.6	1.1	1.4	2.3	1.7	2.6	2.4	1.3	1.0	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	60.9	73.3	79.4	84.1	88.6	91.7	92.1	86.5	80.1	68.0	61.4		
		MCWB	42.2	46.1	51.3	55.4	65.6	68.0	70.6	69.5	66.5	57.2	48.8	45.2		
	2%	DB	48.9	54.2	64.2	72.7	79.1	83.9	87.7	88.1	80.6	73.2	62.0	52.7		
		MCWB	39.0	42.6	47.4	55.0	61.8	66.6	69.5	68.3	64.5	57.7	47.4	41.5		
	5%	DB	43.8	48.6	58.5	68.3	75.3	80.2	84.3	84.6	76.2	67.8	57.1	47.6		
		MCWB	36.5	39.8	45.4	53.6	61.0	65.8	68.4	68.4	63.5	56.1	46.3	39.4		
	10%	DB	39.6	43.8	53.0	63.8	71.8	77.1	80.9	81.4	72.4	63.2	52.0	43.1		
		MCWB	34.6	37.3	43.7	52.0	59.9	64.6	67.1	67.5	62.3	54.3	44.3	37.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	47.2	52.7	59.2	67.8	72.7	73.9	73.5	69.7	61.8	52.2	46.6		
		MCDB	53.5	58.9	69.7	72.8	79.3	82.5	86.5	85.0	80.8	71.5	61.5	56.8		
	2%	WB	39.8	43.3	49.1	56.8	64.9	69.3	71.9	71.5	66.5	59.0	49.4	43.0		
		MCDB	46.7	52.4	61.2	69.2	75.5	79.5	83.5	83.1	76.4	68.7	58.3	50.4		
	5%	WB	37.4	40.5	46.6	54.6	62.3	67.3	70.2	70.1	64.6	57.3	47.1	40.4		
		MCDB	42.7	47.9	55.9	65.3	72.1	77.3	80.4	80.9	73.4	65.7	55.1	45.7		
	10%	WB	35.1	37.8	44.3	52.7	60.5	65.6	68.5	68.5	63.0	55.5	45.0	38.1		
		MCDB	39.0	43.1	51.7	62.3	70.2	74.9	78.1	78.5	70.8	62.1	50.9	42.8		
(35) Mean Daily Temperature Range	MDBR		13.3	14.5	14.6	15.5	15.9	15.9	15.8	16.4	15.9	14.9	14.0	14.0		
	5% DB	MCDBR	20.9	23.3	25.6	23.2	22.2	20.9	20.8	22.0	23.1	23.3	22.9	20.8		
		MCWBR	13.2	13.8	13.5	10.8	11.4	11.0	9.2	9.3	10.9	11.5	12.4	12.5		
	5% WB	MCDBR	18.8	21.5	22.5	21.1	20.1	18.9	18.7	19.7	19.5	20.0	20.2	18.3		
		MCWBR	12.4	13.2	12.9	10.9	11.5	11.0	9.4	9.6	9.9	10.7	11.9	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.379	0.450	0.516	0.480	0.482	0.530	0.523	0.449	0.406	0.354	0.330		
	taud		2.217	2.079	1.938	1.845	2.014	2.065	1.955	1.965	2.135	2.192	2.271	2.248		
	Ebn at Noon		245	251	246	240	254	254	239	236	245	239	237	235		
	Edh at Noon		30	40	53	62	54	52	57	55	43	36	28	26		
(44) All-Sky Solar Radiation	RadAvg		503	777	1022	1358	1651	1866	1856	1703	1308	846	562	429		
	RadStd		67	121	122	180	171	128	149	151	136	112	75	58		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page